

Statistical Report

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Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Can time series models trained on data from 1960–2012 accurately forecast Singapore’s continuing fertility decline over 2013–2024?

TO INCLUDE WHEN INTRODUCTION IS DONE

Include necessary abbreviations when writing it up

4. Justify SARIMA period 12 in Introduction (page 3) The periodogram doesn’t obviously point to period 12, so you need literature to justify it. Add a paragraph in the Introduction discussing the Chinese zodiac Dragon year effect on Singapore births. This is what makes the SARIMA(p,d,q)(P,D,Q)[12] models substantively motivated rather than just data-driven.

Methods

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the two series were used to check for trends, changes in variance and overall appropriateness for ARIMA modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test, residual plots of the series to ensure that the transformation improves the stationarity.

The KPSS test is used for testing whether a given time series is stationary. The time series’ are logged to scale the variance proportionally if there is a trend so to keep relative magnitude of changes in the data in the mind of the model. Differencing is used for the removal of the trends shown inside the data.

Two transformations showed potential as being stationary (first-differenced and logged, and second differenced logged). The TFR data’s KPSS test gave two different p-values for the two differencing settings $p_{\log D=1} = 0.0406$ and $p_{\log D=2} = 0.1$. The logged first difference setting’s p-value was below the 0.05, however, only slightly so it may still be considered if a potential model can produce white-noise residuals. For the TLB data both p-values were over the significance threshold with $p = 0.1$ for both of them meaning that we can retain the null hypothesis, such that the TLB data is stationary with either of the transformations.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set.

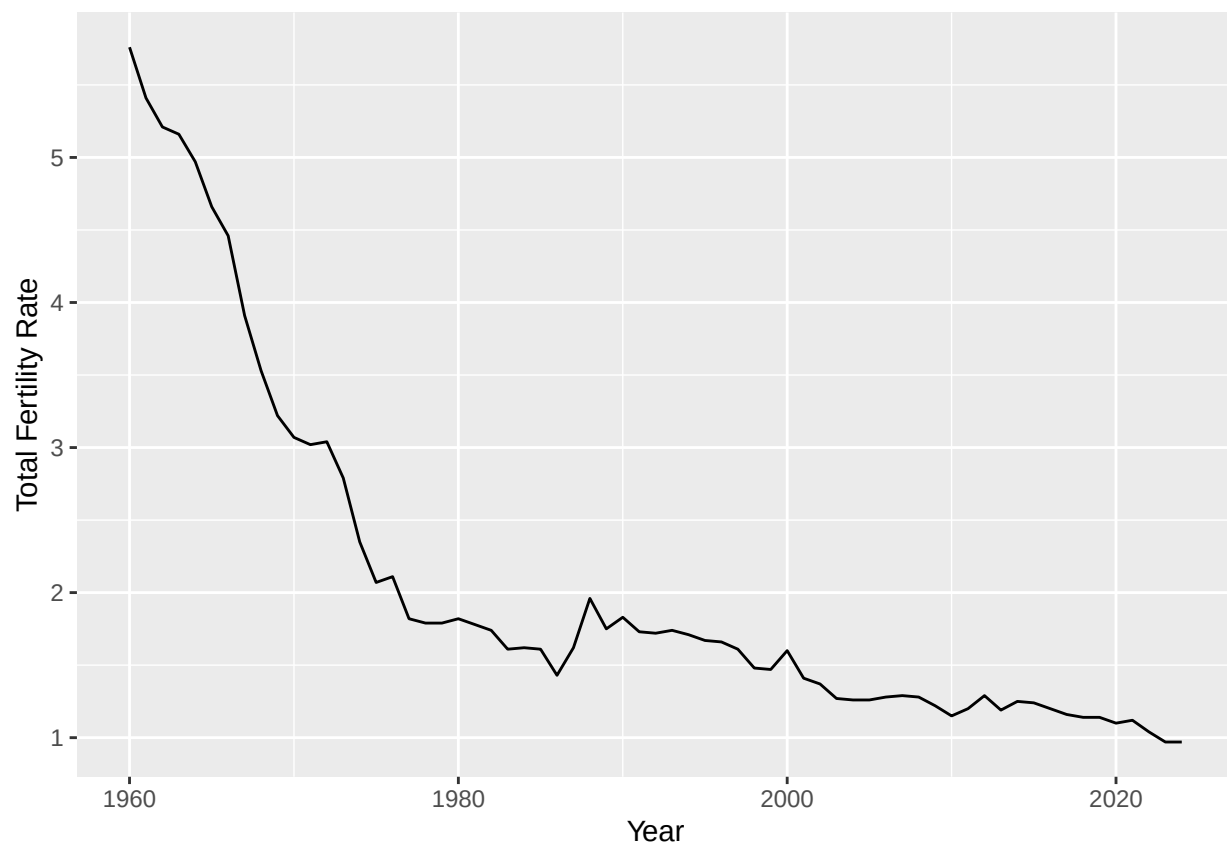


Figure 1: Annual Total Fertility Rate in Singapore

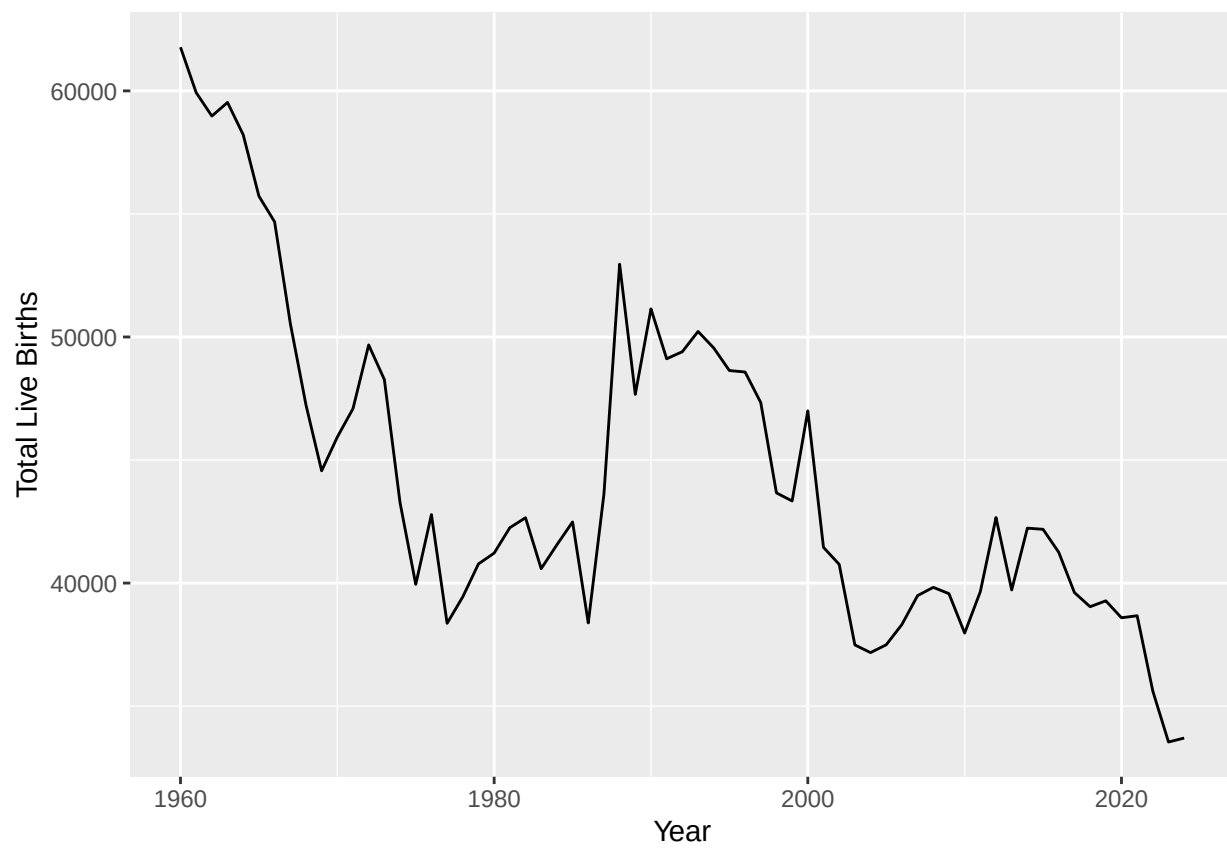


Figure 2: Total Live Births in Singapore

The next step was to pick appropriate models for forecasting for the testing period. ACF and PACF plots were used to see if the time series had hidden correlation which had carried through the transformations. Each time series transformation showed some dependence around lags 11-13 with the most significant spike at lag 12. This potential seasonality made us consider models which would account for this correlation. Models were only considered viable only if their residuals were approximately white noise. Among viable models, the final model selection was based on AIC and forecast accuracy over the 2013-2024 period.

- So informed decision to try ARMIA (12,x, y) and SARIMA models potentially

Series	Models
TFR	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(1,2,1)(1,0,0)[12]
TLB	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(0,2,1)(1,0,0)[12]

Results

Your professor expects the Results section to show:

ACF/PACF plots of the detrended and deseasonalised series
How the first viable model was identified from those plots
How you varied parameters to get additional viable models
How each viable model fits the testing data

Discussion

Conclusion

References

Appendices if applicable.